Three Fermion Generations from E_8: A Machine-Verified Impossibility and Resolution

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We present the first machine-verified resolution of the three-generation problem in grand unified theories. Using the Lean 4 theorem prover [24] with the mathlib library [25], we prove two results across 9 proof files (over 575 verified declarations, zero `sorry` gaps). *Impossibility*: the semi-spinor $\Delta^+(14)$ of Spin(14) decomposes under $\mathrm{SO}(10) \times \mathrm{SO}(4)$ as $(\mathbf{16}, \mathbf{2}) \oplus (\overline{\mathbf{16}}, \mathbf{2})$, giving multiplicity 2 for the **16**. Since $3 \nmid 2$, no intrinsic mechanism in any SO(10+2k) group can produce exactly three generations[MV]. *Resolution*: E_8 admits an $\mathrm{SU}(9)/\mathbb{Z}_3$ decomposition where $248 = 80 + 84 + \overline{84}$ and $\Lambda^3(\mathbb{C}^9) = 84$ and its conjugate $\overline{84}$ together contain exactly three Standard Model generations ($3 \times 16 = 48$ dimensions)[MV]. The number 3 arises as the dimension of the fundamental of $\mathrm{SU}(3)_{\mathrm{family}} \subset \mathrm{SU}(4) \subset \mathrm{SU}(9)$. Both decompositions coexist inside E_8, connected through a common $\mathrm{SU}(7) \times \mathrm{U}(1)$ subgroup. Every dimensional identity is verified by the Lean kernel; every physical interpretation is explicitly tagged.

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INTRODUCTION

The Standard Model of particle physics contains three generations of fermions—an empirical fact without theoretical explanation. In grand unified theories (GUTs), the generation count depends on the decomposition of matter representations under the unification group. The **16**-dimensional chiral spinor of SO(10) [1, 2] accommodates one complete generation[SP]; the question is how many copies appear.

For SO(14), which unifies gauge and gravitational degrees of freedom via $\mathfrak{so}(14) \supset \mathfrak{so}(10) \oplus \mathfrak{so}(1,3)$ [5–7], the semi-spinor $\Delta^+(14)$ is 64-dimensional. Under $\mathrm{SO}(10) \times \mathrm{SO}(4)$, it decomposes with multiplicity 2 for the **16** [8, 19]. The mismatch $3 \nmid 2$ has been known since the 1980s [9–11] but was never formally verified, and the resolution remained open.

We close this loop using interactive theorem proving. In 9 Lean 4 proof files containing over 575 verified declarations with zero `sorry` gaps, we establish:

1. A *universal* impossibility: no SO(10+2k) gives three generations via spinor decomposition, because the family multiplicity 2^{k-1} is never 3[MV].
2. A resolution via E_8: the $\mathrm{SU}(9)/\mathbb{Z}_3$ maximal subgroup provides exactly three generations through the exterior algebra $\Lambda^3(\mathbb{C}^9)$[MV].

To our knowledge, this is the first machine-verified three-generation resolution for any GUT. The methodology—Lean 4 with mathlib—was developed for this project [26–28].

SPINOR PARITY OBSTRUCTION

The semi-spinor of Spin($2n$) has dimension 2^{n-1}. For Spin(14) ($n = 7$): $\dim \Delta^+ = 2^6 = 64$[MV] (`spinor_branching_64`).

Under the maximal subgroup $\mathrm{SO}(10) \times \mathrm{SO}(4) \subset \mathrm{SO}(14)$, the standard spinor branching rule [19] gives

$$\Delta^+(14) = (\mathbf{16}, \mathbf{2}) \oplus (\overline{\mathbf{16}}, \mathbf{2}), \tag{1}$$

where $\mathbf{16} = \Delta^+(10)$ (one SM generation[SP]) and **2** is a spinor of SO(4). The dimension check $16 \times 2 + 16 \times 2 = 64$ is machine-verified[MV] (`branching_rule_dimension`).

The multiplicity of the **16** equals the dimension of the SO(4) spinor:

$$\mathrm{mult}(\mathbf{16}) = \dim \Delta^+(\mathrm{SO}(4)) = 2^{2-1} = 2. \tag{2}$$

This is verified as `multiplicity_is_two`[MV].

The obstruction is now arithmetic. We prove $3 \nmid 2$[MV] (`three_gen_excluded`), which immediately implies that exactly three copies of the **16** cannot be extracted from $\Delta^+(14)$ by any $\mathrm{SO}(10) \times \mathrm{SO}(4)$ decomposition.

Two extensions strengthen the result:

Universality. All embeddings $\mathrm{SO}(10) \times \mathrm{SO}(4) \hookrightarrow \mathrm{SO}(14)$ are conjugate under SO(14), because SO(14) acts transitively on the Grassmannian $\mathrm{Gr}(10, 14)$. The orbit-stabilizer theorem gives $\dim \mathrm{Gr}(10, 14) = 91 - 51 = 40$[MV] (`orbit_stabilizer_check`). No non-standard embedding escapes the obstruction.

No orthogonal escape. For any SO(10+2k), the family multiplicity is 2^{k-1}. We prove that $2^k \neq 3$ for all $k \in \mathbb{N}$[MV] (`no_so_gives_three_families`), since 3 is not a power of 2. The Dirac (128-dimensional) spinor gives multiplicity 4, and $3 \nmid 4$[MV] (`three_gen_excluded_dirac`).

The capstone theorem `spinor_parity_obstruction`[MV] collects all four ingredients—dimension, multiplicity, divisibility, and dimension consistency—into a single conjunction:

$$\Delta^+(14) = 64,\ \mathrm{mult} = 2,\ 3 \nmid 2,\ 16 \cdot 2 + 16 \cdot 2 = 64. \tag{3}$$

The physical consequence is definitive: three fermion generations *cannot* arise from any mechanism intrinsic to SO(14) (or any SO(10+2k)). The generation mechanism must be extrinsic.

THREE GENERATIONS FROM E_8

E_8 provides the extrinsic structure. The chain proceeds in three steps.

Embedding: $\mathrm{SO}(14) \subset \mathrm{SO}(16) \subset E_8$

E_8 has dimension 248 and contains SO(16) as a maximal subgroup[SP]. Under SO(16), the adjoint decomposes as [19]

$$248 = 120 + 128 = \mathrm{ad}(\mathrm{SO}(16)) \oplus S^+(\mathrm{Spin}(16)), \quad (4)$$

verified as `e8_so16_decomposition`[MV]. $\mathrm{SO}(14)\times\mathrm{SO}(2)$ sits block-diagonally inside SO(16): $120 = 91{+}1{+}28$[MV]. The semi-spinor branches as $128 = 64{+}64$[MV]. Together:

$$248 = 91 + 1 + 28 + 64 + 64, \quad (5)$$

verified as `e8_so14_full_decomposition`[MV], with the full chain collected in `e8_embedding_chain_complete`[MV].

$\mathrm{SU}(9)/\mathbb{Z}_3$ decomposition

E_8 admits a *second* maximal subgroup decomposition under $\mathrm{SU}(9)/\mathbb{Z}_3$ [16, 19, 20]:

$$248 = 80 + 84 + \overline{84} = \mathrm{ad}(\mathrm{SU}(9)) \oplus \Lambda^3(\mathbb{C}^9) \oplus \overline{\Lambda^3(\mathbb{C}^9)}, \quad (6)$$

where $80 = 9^2 - 1$ and $84 = \binom{9}{3}$. Both summands are verified: `e8_su9_decomposition`[MV].

The 84 and $\overline{84}$ are conjugate under Hodge duality: $\binom{9}{3} = \binom{9}{6}$[MV]. The $\mathbb{Z}_3$ quotient arises because the center $\mathbb{Z}_9$ of SU(9) acts by cube roots of unity on the three eigenspaces (80, 84, $\overline{84}$)[MV].

Cauchy decomposition and the family triplet

The fundamental $\mathbf{9}$ of SU(9) branches under $\mathrm{SU}(5)\times\mathrm{SU}(4) \subset \mathrm{SU}(9)$ [3, 19] as $\mathbf{9} = (\mathbf{5},\mathbf{1}) \oplus (\mathbf{1},\mathbf{4})$[SP]. The Cauchy formula for exterior powers $\Lambda^k(V \oplus W) = \bigoplus_{i+j=k} \Lambda^i(V) \otimes \Lambda^j(W)$ gives:

$$84 = \underbrace{\binom{5}{3}}_{10} + \underbrace{\binom{5}{2}\binom{4}{1}}_{40} + \underbrace{\binom{5}{1}\binom{4}{2}}_{30} + \underbrace{\binom{4}{3}}_{4}, \quad (7)$$

TABLE I. Generation content from $84 \oplus \overline{84}$ under $\mathrm{SU}(5)\times\mathrm{SU}(3)_{\mathrm{family}}$. The three generations ($3\times 16 = 48$ dimensions) are drawn from *both* $\mathbb{Z}_3$ sectors: the $\mathbf{10}$ and $\mathbf{1}$ from the 84 ($J = +i$ eigenspace), the $\overline{\mathbf{5}}$ from the $\overline{84}$ ($J = -i$). Remaining dimensions ($84 + 84 - 48 = 120$) are exotic. All dimensions are machine-verified[MV].

Rep. $(\mathrm{SU}(5),\mathrm{SU}(3)_f)$	Dim	Source	J	Content
$(\mathbf{10},\mathbf{3})$	30	84	$+i$	$3\times\mathbf{10}$
$(\mathbf{1},\overline{\mathbf{3}})$	3	84	$+i$	$3\times\nu^c$
$(\overline{\mathbf{5}},\overline{\mathbf{3}})$	15	$\overline{84}$	$-i$	$3\times\overline{\mathbf{5}}$
Generation total	48			3×16
Exotic remainder	120			vector-like pairs

verified as `cauchy_decomp_84`[MV]. In representation notation: $84 = (\mathbf{10},\mathbf{1}) \oplus (\mathbf{10},\mathbf{4}) \oplus (\mathbf{5},\mathbf{6}) \oplus (\mathbf{1},\overline{\mathbf{4}})$.

Now break $\mathrm{SU}(4) \to \mathrm{SU}(3)_{\mathrm{family}} \times \mathrm{U}(1)$[CP], under which $\mathbf{4} = \mathbf{3} + \mathbf{1}$. The $(\mathbf{10},\mathbf{4})$ piece branches as $(\mathbf{10},\mathbf{3}) \oplus (\mathbf{10},\mathbf{1})$, giving *three copies* of the $\mathbf{10}$ of SU(5): $\dim = 10\times 3 = 30$[MV] (`three_generation_count`).

Similarly, $(\mathbf{5},\mathbf{6}) \to (\mathbf{5},\mathbf{3}) \oplus (\mathbf{5},\overline{\mathbf{3}})$ and $(\mathbf{1},\overline{\mathbf{4}}) \to (\mathbf{1},\overline{\mathbf{3}}) \oplus (\mathbf{1},\mathbf{1})$. The conjugate $\overline{84}$ decomposes analogously; crucially, $(\overline{\mathbf{5}},\overline{\mathbf{3}})$ with dimension 15 appears in the $\overline{84}$, providing $3\times\overline{\mathbf{5}}$. Table I collects the generation content from both sectors. Three complete SM generations contribute $3\times(\mathbf{10} + \overline{\mathbf{5}} + \mathbf{1}) = 3\times 16 = 48$ dimensions[MV] (`three_gen_dim`, `generation_matter_total`), drawn from both the 84 and $\overline{84}$. The exotic content is $84 + 84 - 48 = 120$ dimensions, forming vector-like pairs that can acquire GUT-scale masses[CP].

The eight-part capstone `e8_three_generation_theorem`[MV] collects: $\binom{9}{3} = 84$, $10\times 3 = 30$, $5\times 3 = 15$, $10 + 5 + 1 = 16$, $3\times 16 = 48$, $84 - 48 = 36$, $48 + 36 = 84$, and $\dim(\text{fund. SU}(3)) = 3$. The anomaly-free eigenspace theorem `j_sector_anomaly_free`[MV] additionally verifies $\mathrm{Tr}(Y) = 0$ for each J eigenspace independently.

Compatibility of the two decompositions

The SO(16) and SU(9) decompositions of E_8 share a common subgroup $\mathrm{SU}(7)\times\mathrm{U}(1)$ (dimension $48 + 1 = 49$[MV]). The 91 generators of SO(14) distribute across the SU(9) sectors as:

$$91 = \underbrace{49}_{\text{in } 80} + \underbrace{21}_{\text{in } 84} + \underbrace{21}_{\text{in } \overline{84}}, \quad (8)$$

where $21 = \binom{7}{2} = \dim \Lambda^2(\mathbb{C}^7)$[MV]. This is verified as `so14_compatible_split`[MV]. Every generator of SO(14) is accounted for in both decompositions.

The complete argument is collected in the seven-part `three_generation_theorem`[MV], which combines:

(a) impossibility at SO(14) ($3 \nmid 2$), (b) embedding in E_8 ($91 + 1 + 28 + 64 + 64 = 248$), (c) SU(9) decomposition ($80 + 84 + 84 = 248$), (d) three generations ($3 \times 16 = 48 \leq 84$), (e) exotics ($84 - 48 = 36$), (f) compatibility ($49 + 21 + 21 = 91$), and (g) the origin of 3 ($3 \mid 3$ and $3 \nmid 2$).

DISCUSSION

Epistemological boundary. Every dimensional identity in this Letter is machine-verified ([MV]): over 575 verified declarations across 9 Lean 4 files, zero `sorry` gaps. The arithmetic $\binom{9}{3} = 84$, the branching $84 = 10+40+30+4$, and the generation count $10 \times 3 = 30$ are mathematical facts accepted by the Lean kernel.

The identification of $\mathrm{SU}(3)_{\mathrm{family}}$ as the *physical* family symmetry is candidate physics[CP]. The claim that exotics decouple at the GUT scale is candidate physics[CP]. All Yukawa textures and proton decay predictions from the E_8 breaking chain are open problems[OP]. We tag every claim explicitly.

Chirality boundary. The generation mechanism verified above is algebraic: it holds for *all* real forms of E_8, including the physically relevant $E_8(-24)$ containing Spin(3,11). The Clifford periodicity distinguishes signatures—$\mathrm{Cl}(14,0)$ has $p-q \bmod 8 = 6$ while $\mathrm{Cl}(11,3)$ has $p-q \bmod 8 = 0$—but the generation count $3 \times 16 = 48$ is signature-independent[MV]. Distler and Garibaldi [22] proved that no embedding of the Standard Model in any real form of E_8 can produce three *massless chiral* generations from the adjoint. Wilson [23] argues that a compact gauge group is intrinsically massless, so for theories with explicitly non-zero masses—such as $E_8(-24)$, which contains $\mathrm{SU}(2,2)$ rather than the Poincaré group—the Distler–Garibaldi no-go theorem does not apply. The non-compact degrees of freedom serve as mass parameters rather than ghosts. Which definition of chirality is physically correct remains open[OP], but the *algebraic* question can be resolved.

We provide a two-level algebraic definition. *Level 1* (Definition B): $\Lambda^3(\mathbb{C}^9)$ is *not* self-conjugate as an SU(9) representation, since self-conjugacy of $\Lambda^k(\mathbb{C}^n)$ requires $2k = n$, and $2 \times 3 = 6 \neq 9$[MV] (`wilson_chirality`). Wilson [23] makes this argument explicitly: "Since $\Lambda^3(\mathbb{C}^9)$ is not the same as the antisymmetric cube of 9^*, the theory is chiral." Because 9 is odd, *every* exterior power $\Lambda^k(\mathbb{C}^9)$ for $0 < k < 9$ is non-self-conjugate[MV] (`odd_nine_all_chiral`); the chirality of SU(9) is structurally robust. *Level 2* (Definition D): the A_8/D_8 overlap matrix—computed from the non-nested maximal-rank subalgebras SU(9) and SO(16) of E_8—quantifies the asymmetry[CO]: the 84 roots of sector 1 split as $28 + 35 + 21$ under $D_8 \supset D_7$ chirality (S^+ vs. S^-), giving a chirality index $\chi = 35-21 = 14$[MV] (`chirality_index_sector1`), while the total 248 remains balanced ($64^+ = 64^-$)[MV], consistent with Distler–Garibaldi. The combined definition is collected in `massive_chirality_definition`[MV] (12-part conjunction) and the full certificate `complete_massive_chirality_skeleton`[MV] (20-part conjunction), documented in `massive_chirality_definition.lean`, `e8_chirality_boundary.lean`, and `exterior_cube_chirality.lean`.

Anomaly cancellation and the $\mathbb{Z}_3$ sector operator. The SU(5) anomaly cancels generation by generation [4, 21]: $A(\mathbf{10}) + A(\overline{\mathbf{5}}) = 1 + (-1) = 0$[SP]. For three generations: $3 \times 0 = 0$[MV] (`anomaly_three_generations`). The $\mathbb{Z}_3$ automorphism σ of E_8 that defines the $\mathrm{SU}(9)/\mathbb{Z}_3$ decomposition induces a complex structure $J = (2/\sqrt{3})(\sigma + \frac{1}{2}\mathrm{Id})|_V$ on the 168-dimensional matter space $V = 84 \oplus \overline{84}$, with $J^2 = -\mathrm{Id}$[CO]. Crucially, J does *not* reduce to the chirality operator γ_5[CO]: one Standard Model generation straddles both J eigenspaces, with the $\mathbf{10} + \mathbf{1}$ (containing Q_L, u^c, e^c, ν^c) in the $+i$ eigenspace and the $\overline{\mathbf{5}}$ (containing d^c, L) in the $-i$ eigenspace. Instead, J encodes the SU(5) multiplet structure: it is the algebraic reason why fermions are grouped into the $\mathbf{10}$ and $\overline{\mathbf{5}}$. Each J eigenspace is independently anomaly-free: $\mathrm{Tr}(Y) = 0$ for $\{\mathbf{10}, \mathbf{1}\}$ and for $\{\overline{\mathbf{5}}\}$ separately[MV] (`j_sector_anomaly_free`). This "miraculous" cancellation—previously an unexplained feature of the Standard Model—becomes a consequence of σ preserving the algebraic structure of E_8[CP].

Why E_8 succeeds where SO(14) *fails.* The root cause is algebraic: spinor representations of orthogonal groups yield multiplicities 2^{k-1}, which are always powers of 2. The exterior algebra $\Lambda^k(\mathbb{C}^N)$ has no such constraint—$\binom{N}{k}$ can be any integer. In particular, $\binom{3}{1} = 3$[MV], while no $2^k = 3$[MV] (`three_not_power_of_two`). The transition from SO(14) to E_8 replaces the SO(4) family factor (spinor dim 2) with $\mathrm{SU}(3)_{\mathrm{family}}$ (fundamental dim 3). The extra structure has dimension $\dim \mathrm{SU}(4) - \dim \mathrm{SO}(4) = 15 - 6 = 9$[MV], housing exactly SU(3) ($\dim 8$) + U(1) ($\dim 1$)[MV].

Comparison with prior work. Wilson [13, 14] identified discrete order-3 elements of E_8 whose centralizer is $\mathrm{SU}(9)/\mathbb{Z}_3$, providing a fundamentally discrete three-generation mechanism. Our contribution is to *machine-verify* the complete dimensional chain from impossibility through resolution. Our chirality boundary analysis, informed by Distler and Garibaldi [22], clarifies the precise scope of Wilson's escape: the algebraic prerequisites are verified, while the chirality definition remains open[OP]. Calabi-Yau compactification [18] obtains three generations from the Euler characteristic $|\chi|/2 = 3$ of specific manifolds, but this is geometric, not algebraic. BenTov and Zee [17] explored SO(18) and showed that the family problem persists for all SO($10+2k$); our `no_so_gives_three_families`[MV] proves this uni-

versally. Bars and Günaydin [12] were the first to use E_8 for grand unification; our work adds formal verification.

Exotics. The 120 exotic dimensions in $84 \oplus \overline{84}$ (Table I) form vector-like pairs under SU(5) that can acquire GUT-scale masses without breaking the Standard Model gauge symmetry[CP]. The ratio of SM matter to total ($48/168 \approx 29\%$) is typical of E_8 models [15].

CONCLUSION

We have proved, with machine verification, that three fermion generations cannot arise from any orthogonal group SO(10+2k) via spinor decomposition—a universal impossibility—and that E_8 resolves this obstruction through the $\mathrm{SU}(9)/\mathbb{Z}_3$ decomposition, where $\Lambda^3(\mathbb{C}^9)$ naturally contains $3 \times 16 = 48$ dimensions of Standard Model matter. We further provide a two-level algebraic definition of chirality: non-self-conjugacy of $\Lambda^3(\mathbb{C}^9)$ (Definition B[MV]), quantified by a chirality index $\chi = 14$ per $\mathbb{Z}_3$ sector (Definition D[MV]), compatible with the Distler–Garibaldi no-go theorem. The physical observable corresponding to massive chirality remains open[OP].

The number 3 is not put in by hand. It is the dimension of the fundamental representation of $\mathrm{SU}(3)_{\mathrm{family}}$, which emerges from the chain $E_8 \supset \mathrm{SU}(9) \supset \mathrm{SU}(5) \times \mathrm{SU}(4) \supset \mathrm{SU}(5) \times \mathrm{SU}(3)_{\mathrm{family}} \times \mathrm{U}(1)$.

The complete argument—9 proof files, over 575 verified declarations, zero `sorry` gaps—is available from the corresponding author upon reasonable request [29]. We propose that machine verification become standard methodology for GUT algebraic constructions, eliminating the class of errors that can persist undetected for decades.

The machine verification was performed using Lean 4 and the mathlib library. The author thanks the Lean and mathlib communities for providing the infrastructure that made this work possible. R. A. Wilson's work on discrete order-3 elements of E_8 and the $\mathrm{SU}(9)/\mathbb{Z}_3$ decomposition provided essential mathematical foundations; correspondence with Wilson clarified the chirality boundary. Proof engineering and manuscript preparation were assisted by Claude (Anthropic); all mathematical content, research direction, and physical interpretation are the sole responsibility of the author.

DATA AVAILABILITY

The Lean 4 source code for all machine-verified proofs is available from the corresponding author upon reasonable request [29].

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